

ActInf Textbook Group ~ Cohort 9 ~ Session 7

(Chapter 5) ~ 10/2/2025

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Okay, welcome everyone. Uh, thanks for attending today the active inference institute's textbook reading group. It is October 2nd uh doing the relative to the western hemisphere the evening session of the textbook reading group. This is cohort 9 reading chapter five of the textbook. So, we're wrapping up uh essentially the first half of the book, and this chapter is a bit of an outlier relative to what we've seen so far. It does follow from and connect to the preceding chapters, but it's here that we get much more deeply into I would say the more or less kind of like neurobiological underpinnings of active inference beyond just looking at notions of generative models and free energy as kind of a formalism involving gradient descent and um sort of bayesian statistics and mechanics. um you know it comes into question well you know this this is all much of this is derived from neuroscience and so um even though we get a lot of these cognitive concepts of perception and action and uh and the rest sensory attenuation it's here that we start to see something much closer to uh an understanding of maybe how neuronal populations communicate with one another what these message passing schemes excuse me, all about. So this um this chapter begins with an interesting quote uh from Rulo Lenus, Colombian neuroscientist uh quite a reputation behind him has worked with uh NYU School of Medicine. Uh basically there are two types of animals. Animals and animals that have no brains. They are called plants. They don't need a nervous system because they don't move actively. They don't pull up their roots and run into forest fire. Anything that moves actively requires a nervous system. Otherwise, it would come to a quick death. It's a very interesting way of opening up this chapter. And from my personal point of view, I think it is emphasizing the significance of having a nervous system and sort of neuronal communication happening within an organism. Um that said, using the active inference framework, one could nonetheless model uh in fact a plant whether it be in the midst of a forest fire or not. Um that depends on how you want to draw the marov bank blanket. What are the statistical boundaries separating the plant from the not plant or environment

around it and what are the sort of inputs or observations uh received by the plant versus what are its outputs. Uh so we as humans know it's very important to us to have plants for uh you know successful production of oxygen. uh so they do have outputs and there are different things that happen that lead to plants being able to persist or not. Uh so yeah I I find this quote very interesting but I I think that there need to be a few asterisks after it um maybe to kind of understand it and situate it here. Nonetheless, the kind of the the broader theme beyond just neurobiology is understanding active inference and its attendant process theory or neuronal process theory. Um we know that the the free energy principle is a principle. Um so here a process theory refers to how principles like the free energy principle might be implemented in certain systems like the brain and this allows us to develop hypotheses answerable to empirical data and further produces a dialogue between say technical audiences like engineers or data scientists uh as well as neurobiologists uh cognitive scientists uh essentially kind of um kind of producing the dialogue that allows us to look at these math complex mathematical formulisms, look at empirical data uh and and what we know about the brain and seeing where these two things match up. Um in active inference the basian belief updating we've covered so far chapters 1 through four may be carried out via here basian message passing for there are different kind of algorithms or forms that message passing can occur. Um all the ones in active inference we look at is be in the kind of variational category um uh one that's very commonly used is uh marginal message passing is what it's called. There's a nice paper on that by Par and Fristen and others for 2019 um where marginal message passing is kind of takes the best of both world worlds as it were from variational message passing which at times can sort of lead to um overconfidence in its outputs um versus belief propagation which uh those who engage in data science are um often well aware ware of and it's uh the issue with belief propagation is just that it's it's very computationally demanding um and may lead to intractability problems and so we know from reading prior chapters um a big part of active inference and the free energy principle involves these variational methods allow for overcoming those kinds of hurdles. So um I would recommend uh you know further reading for those who are interested in the kind of the specificities of different kinds of message passing. Um but here we'll just we'll just kind of focus on um kind of the way that these things are described. Um message passing what is that? Um neurons pass messages to one another? Do all me do all neurons pass messages to all other neurons? that is are they all fully connected up to one another? Uh absolutely not. Uh there there's much more sparity sparse connectivity in the brain. Certain neurons connect to certain other neurons for those familiar with Hebian learning and synapses.

There are neurons who were never communicating with one another who then start to communicate with one another based on their relative activity and who they're connected to. Um, and so whenever we do these kinds of, you know, neuroimaging experiments or or or try to create an active inference model of these things, of course, we're we're we're typically not trying to model billions and billions of neurons. We're trying to um collect information for neuroimaging and empirical data. We're collecting uh kind of signals from entire populations or congregations of neurons that are supposed to be more or less within certain areas of the brain. So if you're looking at like um you know the left temporal lobe uh then then you're you're commonly like for EEG you're putting an electrode there and there's a singular signal and you're assuming that that more or less is a proxy for all the communication going on somewhere within that area of the temporal lobe. And so now if you're going to do a study where you want to understand the temporal lobe, you're not looking at millions of neurons in the temporal lobes individually. you're looking at a whole population. So that kind of idea of understanding neuronal populations and how they communicate with one another, one is obviously much more efficient um in the sense of you don't have to model all of these individual neurons um and then two the idea of do neurons who are together kind of communicating that way do they tend to have certain kinds of behaviors or lead to larger emergent behaviors from the the organism. So that that's kind of what allows us to draw out these sort of um these diagrams that we've seen so far. It's like well why do we only have five circles or variables? Why don't we have a billion? Um that's why uh in a sense. So um with this chapter we quickly get into sort of a lot of technical information. This is what is known as a canonical m cortical microcircuit as it's quite obvious from looking at the chapter. Um here the idea is there's communication between these different kinds of neuronal populations that might be able to be located in certain areas of the brain. For example, on the far left we see that there's an attempt to sort of map uh to these neuron populations to different areas of the brain. the basal ganglia secondary thalamus. Um another um something that's that's quite significant in uh sort of empirical neuroscientific and neurobiological studies is the idea of looking at um pyramidal cells. Um so that we also have different types of neurons right so they don't all necessarily function similarly. The main thing that I would say anyone reading this chapter could potentially take away whether they want to really get into the fine details of neurobiology or not. is that one of the core kind of empirical findings is that deep pyramidal cells, that's what they're called, um that are kind of uh deeper in the brain as it were, uh send these projections or expectations uh down to superficial pyramidal cells that are much

closer to um the other end of the layers here. uh that is ones that are closer to sensory receptors. So you can imagine like um if we had a spectrum or or or kind of like this kind of layering of these cortical layers, some of them are much closer to your eyes so to speak. That is your eyes taking in visual information, sensory information. It it will that information kind of hits so to speak. it kind of touches on or is received as an input at these kind of more shallow layers and then as you go um that then go uh towards the deeper layers, right? So you have this kind of communication back and forth uh depending on depth, right? So these superficial more shallow layer neurons send prediction errors deeper. Um so you have these bottom up prediction errors. Then the deeper neurons or layers send down these kind of top down projections or predictions. So what does that mean? Well, you can think of someone who, you know, believes that it's daytime right now and then they step outside and they see that there's no sun and they check the time and it's midnight and they suddenly realize, oh, it must not be daytime. it must be nighttime. So they kind of figuratively you could say that at the deeper layers they had this belief that it was daytime in fact but then by receiving this incoming sensory information um you kind of had the the sensory observation come into conflict with the top down expectation or projection the belief and then there was a correction to that belief. Right? And so this is sort of the this is sort of the arena of belief as it were where where prediction errors are identified sent uh deeper into the sort of the layer uh deeper into the cortical layer structure here and then kind of meet with the the expectations that were already there in the brain and uh there's some kind of presumably resolution that happens. How does that happen through all of this sort of message passing right? So you have a belief that belief gets corrected by sensory information. Um so this is the the general sort of scheme for that. But to for those who are like why why do we have three different um illustrations here? This is simply like three different ways of kind of communicating the same thing. Uh for those who have already skipped ahead to like chapters seven and eight recognize the difference between discrete versus continuous time models, uh the this illustration to the far right will look a little bit more familiar. We have our S's for hidden states. We have our O's for observations, right? We have our our G uh whenever we compute expected free energy per policy. We have our A matrix for perception. Uh our likelihood matrix. So you know the that's what is it day or night those are the hidden states s uh you know do I see sunlight and do I see the time you know being uh you know yes or no those are observations it's in the a matrix that they kind of touch uh there um yeah Alexander go for it >> um so this may just be me being very literal with the terms because of the neurobiology background But um based on what you

described about the cortical layer and like the relationship between the deep parameal and the superficial parameal neurons um it seems like and tell me if this is correct or not. Um that the the understanding of this circuit is that the deep parameal neurons um are already have a prediction or are already doing a thing in in the brain. Um, and that the incoming input into layer 4 um is able to update or sort of like redirect, but even in the absence of layer 4 input, they would still be doing something. >> Yes. submission. >> Yeah. Yeah. I think that I think that's fair to say. Um yeah, it's it's in the sense that so long as you have the message passing occurring to use that language or the the idea that these neurons are in fact, you know, linked to one another in some fashion and are actively sending out signals. It's like, you know, at no point, as far as I'm aware, you I'm uh I mean, I I personally don't have a neurobiology background, so I'd almost be willing to take your word on a few aspects of this, but having spent a lot of time with this chapter and and then also the kind of the mathematical formalisms and the rest like that, I am strongly familiar with, it's like, yeah, as long as message passing is occurring, I mean, these neurons don't like kind of shut down, right? I mean, they're they're all sort of like living in a in a continuously kind of living and and operating brain. So, so yeah, that that'd be right. Um, and it's it's it's sort of like, you know, there's communication from the superficial cells to the deep, from the deep to the superficial. And so there's this kind of updating that happens between them, right? They sort of meet in the middle in different sorts of ways. Um so if one is willing to accept the sort of formalisms and and hypotheses of active inference and the free energy principle uh then then we would not only say that this happens but we would actually say that it happens in a particular way and in that way would sort of look like this. It would look like there are prediction errors which is not unique to active inference. Um it's rather that there would be this kind of this kind of basium uh basian paradigm that we're in and you know a lot of the specific functions and formulas and whenever we look at you know the maths behind free energy and expected free energy we from a active inference standpoint we would be saying like yes this is how that happens. Um but it's it's it seems pretty uh widely accepted through many areas of neuroscience. People who are you know less interested in some of the things that active inference looks at but um you know those who look at uh dopamine you know and are totally centered on on neurobiology or neurotrans transmitters. And it's kind of widely accepted that dopamine encodes uh reward prediction errors. And so this idea of there being prediction errors kind of assumes that there's something already present and operating whether it be beliefs or whatever word we want to use there. um those are sort of already functioning in the brain perhaps at the deeper areas that

the deeper uh parameal cells and then with new sensory inputs there's some kind of you know error that happens between the beliefs that were held versus the sensory information that was received. Um and one more question about um some of the language. Uh you said at one point that the projections from the neurons um from one region to another are like beliefs. Is that like a a metaphor or is that like literally how to interpret um a kind of circuit? is that um the existence of connections or message passing is the propagation of a belief. >> I see. Yeah. Yeah. It's it's kind of like yes to everything. Um I would say so unfortunately the textbook kind of switches around a little bit with the terminology which is why even mids sentence I'm like projections expectations like before I continue um like as far as mathematical formalisms go like here this this middle one is for the the continuous case so it's just it's very like looking at this from a mathematical perspective this μ here refers to an um ultimately like an average or or an expectation in the mathematical sense. Um and so that that term can be like very reasonably accurately used here like an expectation. Um a projection is a fair term to use which has also been used in the textbook at certain points. Um projection kind of implies that. Um, but it also implies like that, you know, some kind of movement or pushing like we're we're not we don't just have this expectation but we're moving it is does that mean we're moving it as a message to these other cortical layers or does it mean that you know it's being projected into the future like this is what I expect will happen in the future little ambiguous both of them could be reasonably said to be uh valid here though and then as far as calling it a belief. Um, that's a bit tricky too in the sense that I would certainly not say belief in a kind of colloquial everyday sense of like we can explicitly, you know, quote unquote rationally talk about what our beliefs are. Like I believe, you know, such and such like I'm not sure if my arm believes that it's exactly here or exactly there, but there's something going on to where it like my brain is locating where my arm is. uh you know so that the idea of like you know many of these processes are are largely unconscious so I I wouldn't say that um you know belief in a very everyday kind of sense but um given that this expectation is coming from a place that you know hasn't even necessarily received sensory input yet on the current situation in a sense it is like a belief um and then finally like belief propagation as a mathematical formalism. Um, in that case, it's just we're just kind of moving these numbers around. We're taking a an expectation, an average, and we're we're kind of pushing it as a message to, you know, some other structure that will then do something with that. Use it to compute its own average or something. So, kind of like playing telephone. Um, I hope that was useful. I'm sorry that I couldn't give like a more like like precise answer, but it's just uh yeah, it's a big picture. Cool.

Um yeah, that's again this is just another way of kind of formalizing the figure we already saw. It's just kind of drawing it from left to right. um in the sense that you know there are there's communication sort of vertically and horizontally as it were um between these neuronal populations. So we can see there there's descending prediction error. There's us uh excuse me descending predictions and ascending prediction errors like I already mentioned. Um and then you can see the communication between specifically like deep and superficial. um for sake of time I'll I'll just say that this is a kind of another way of of representing that um and then uh motor commands. So they they they you know not everyone who comes in active inference is specifically interested in say proprioception. uh like as as we might be familiar with by now at this point in the textbook, there's sort of three different uh forms of of of inference that can occur uh or or sensory data that can be collected that being extterosceptive that is what's from outside of the agent or organism uh you know so that that would relate most commonly to the five senses what I see what I touch what I feel um externally um then there's interosceptive which would be the the body's ability to sort of read itself internally. So what does it mean for the brain whenever it notices that its heart rate has you know risen and uh you know its stomach is growling and uh you know these sort of internal signals that are received um that we don't use our eyes and our outer like dermal like our skin to feel and uh you know the other five senses that we usually think of as you know receiving in information from the outside. Then there's proprioception which largely just has to do with movement um generally kind of motor commands. Um they don't they they tend to be kind of distinguished in that they they have to do a lot more with sort of um reflex arcs. They have to do with how you know how does um the forearm rotate on its joint. um how does it do that in a way that you know can can um sort of be inferred so that the organism can actually move its body and its joints in a way that it wants to whether consciously unconsciously or otherwise um most importantly I think here is that we're looking at this notion of like okay in order for my hand to go from point A to point B or like from a resting state to picking up my glass so that I can drink my water. Um, a lot of inference sort of needs to to happen, right, to get from point A to point B. Um, there's a lot of prediction error that needs to be sort of um, computed and then through action overcome. It's like, well, if I want to pick up my water, but my hand is over here in a resting state, I'm already at a a pretty strong prediction error uh, relative to this preference or desire that I have for picking up the glass of water, right? So, I need to kind of minimize that. Uh, presumably by moving my hand closer and closer to picking up the glass of water. Um, that means I have to move my arm. That means I have to engage the the muscles in my arm in

order to like actually move it. Um, so there's just sort of a variety of processes here that are occurring down to like a rather micro sort of level. Um, and this is one way of attempting to sort of communicate how that occurs in the formalisms of active inference where we do have these these descending expectations or projections or beliefs. Um and then we have uh through the motor neurons and like actual interaction with the reality of receiving the proprioceptive inputs. they have a prediction error that can then be computed from the incoming observations that are received and then they attempt to sort of map this to neurobiological correlates by placing it on top of this um sort of implicit diagram with the dorsal and ventral horns subcortical structures. This is I they go most deeply into talking about action here. Um kind of the the basal ganglia circuit in its relationship to neuronal message passing. Again, this is um this is kind of like being able to say like, okay, well, we have we have this whole kind of reactive style active inference. We're we're minimizing variational free energy. um we we we try to increase accuracy of our model. We try to reduce complexity. Um but then why is it that we have expected free energy and expected you know energy of the future? Um and the idea of having goals and trying to accomplish goals and so on. And so this kind of gives a little bit more, you know, biological evidence to that and that we we we have references to um you know, sort of how the brain decides to take on particular actions not just to uh increase accuracy and reduce complexity of its model, but also uh and in tandem sort of with that to achieve um goals. And then these are just further ways of kind of talking about that because action uh can actually be sort of divided up whenever you get deeper. There's a really nice concise formalism for action or policy inference and active inference that I've really enjoyed showing during like talks I've given on these things um that unfortunately they don't have in this book. Uh it's it's used throughout probably 90% of all the simulations that this book refers to. It's used in the SPM library, the PIMDP library, RX and uh I'm not sure it was RX infer but many of these other libraries. Um but essentially it the policy inference boils down to um sort of computing expected free energy to which relates to kind of like deliberate decision making or trying to achieve particular goals like oh I'm moving my hand so that I can pick up you know this glass of water so that I can drink it. um and that that relates to these sort of direct pathways um along the the the ganglionic circuitry here. Meanwhile, um we also have habits, right? Uh which in a colloquial sense many people are familiar with. Um you know, whether it be a sort of a bad habit, chewing fingernails, smoking, uh you know, a cigarette, uh or otherwise. Um or just kind of, you know, habitual ticks or, you know, knocking on wood, whether you believe in the idea of doing that or not, but you've done it so many times that you

can't help but not do it. Um so that the the idea of having habits that you know in the in the formalism here it's kind of just through repetition one builds up a habit. Um and then whenever it comes to sort of neuromodulation and neurotransmission, uh whenever someone is not the the hypothesis here is that whenever someone is not terribly confident in their ability to take actions that will successfully lead to accomplishing their goals, whether it be picking up the glass of water or it be making some major, you know, financial decision. for their livelihood and their their future or whatever it be. Um, when lacking confidence, um, there's kind of an activation of other areas of the brain that relate to habits, which would sort of be like contextfree policies or actions. It's like, oh, whenever I don't know what to do, this is what I do. Um, unconsciously, not not whatsoever necessarily meaning that you're, you know, consciously doing this. Um so that gives us E which is a prior on policies or a prior on actions. So you can also think of that in a sense is like um you know itself being a sort of expectation of what to do regardless of circumstance. um with some of my my computational psychiatry work that I've been doing with a a group at the institute recently um like one way of viewing an individual who you know unfortunately has been through some form of um trauma or struggles perhaps with certain uh kind of kinds or veillances of anxiety. Um it can be the case that over a very significant amount of time whenever someone is struggling with those kinds of um um symptoms as it were uh they they tend towards kind of avoidant behaviors in order to avoid uh triggering or or uh otherwise anxietyinducing situations. And there's an interesting hypothesis there that by doing that more and more even as someone continues to you know say take a prescribed medication seek additional care practice emo emotional regulation strategies see a therapist or otherwise once that habit is built up strongly enough uh one can become very used to kind of engaging in those avoidant behavior behaviors um despite doing otherwise overall like really well um you mentally with their health. And it's it's tough to break habits. In order to kind of break a habit, the hypothesis here would be that you would need to repeatedly engage with something that is mutually exclusive to the the habit, right? Um, so we can think of someone who who has been smoking, you know, cigarettes for a very long time or something. It's like in order to quit once you get past the the the chemical addiction, um, you would need to repeatedly not smoke in order to also relieve this sort of mental habit that this this the activation of this indirect pathway that says, "Oh, I'm not sure what to do or maybe I'm stressed. I will go smoke a cigarette." is a contextfree you know way of dealing with this situation you know whether is it the morning you can smoke is it the afternoon you can smoke is it the evening you can smoke that's the sense of the context for am I on break right now I

can smoke like right so that's um it it says a lot about animal behavior too right because whenever we we consider the idea of like oh you know animals including human infants being born with different kinds of like sort of emergent behaviors that we frequently see despite that the the the child or the the infant or the animal never actually being taught how to um do those things or being taught to you know an infant not being taught to cry rather they tend to do it on their own. Um it's sort of like we could almost formalize that as a kind of ingrained habit that that sort of pre-exists. Um well, it's through repetitions of of action and turning perception that perhaps a baby eventually learns to move from the crying to an alternative which might be using you know language once they hit a certain age or and are able to to articulate themselves and you know kind of perceive that there's a meaning between this the the sounds of certain words or certain sounds and them having a reference. of saying you're hungry means something um to your to your parent. Uh right. So um with all of that with talking now about sort of neuromodulation and transmission the idea of confidence um whenever I used the word confidence earlier um there's sort of technical formalism of that here is expected policy precision. Um essentially when we think of neurotransmitters whether you're an active inference or not we tend to view them as they have an impact they they modulate um communication between neurons. So that would mean that they'd have a direct impact on these message passing schemes that are going on in the brain. um most simply we could consider them as sort of um amplifying or or sort of dampening different signals. Um so whenever we think of uh a likelihood we could say you know one's um belief in seeing the sun that means it's daytime. Uh they don't see any sunlight that means it's nighttime. You could say that they have that as a sort of 2x2 matrix. Um you know it's its probability of sun goes to day is a 1.0. That means you're completely certain of that. Um and then the opposite you don't see the sunlight it must be nighttime. That's a 1.0 right and with with probability summing to one that means that seeing the sun and it being nighttime has a probability of zero. $1 - 0$ is one everything sums to one and vice versa. Um if we reduce the precision on that likelihood uh on that overall matrix say to then what happens is that now I now it's as if I believe that if I see the sun there's about a $\frac{3}{4}$ of probability that it's that it's daytime and then a one quarter probability that oh no actually I'm seeing the sun and it's night like is it unlikely Sure. But is it like impossible according to my new downtuned likelihood matrix? No. It's a it's about you know 0.25 probable. Um so that's sort of the the notion of precision and it that becomes really significant in a lot of different kinds of cognitive phenomena. Um for example, whenever it comes to the study of sort of hallucination and delusion, um it might be the case that you could

formalize that as there's reduced precision on between what one believes and what one sees. You know, that is between one's inferred hidden states of the world versus the sensory observations that are actually uh received. um by down tuning that precision it's almost like you're saying oh I'm less certain that what I'm seeing matches up with what I would you know believe or I'm I'm it's it's more difficult to say right like what would happen if we just go back to the simplified example I gave a moment ago of the 2x2 matrix and say oh if I see the sunlight it's a .5 probability that it's daytime probabilities sum to one so that would mean that the probability of it being nighttime time would also be 0.5. So that means I literally I see the sunlight, but based on my likelihood matrix being downtuned to precision of 0.5, what's happening is I have no idea if this me seeing the sun means it's day or night. It's no longer a useful sensory signal. That's clear. It's like this isn't telling me to think one way or the other. Now, um that that could lead to some some major major issues, right? Not just for someone trying to figure out if it's day or night time, which already has its own issues. If you're trying to determine like is a sort of horistic like, oh, do I should I be sleeping or should I be getting ready to go to work? Um is it day or night? I just woke up. Uh but it could also be, you know, for for much higher stake situations, right? to have to kind of have a downtuned um you know understanding of the world and and it opens up doors for potential like many to many mappings uh to where it's very difficult to sort of parse through what different things mean and being able to actually use sensory information in any kind of useful way for for navigating the world. Um and it's there that you know the you know um delusion is a is a general sense of you know despite receiving particular kinds of evidence that maybe an external observer might say oh yeah no if you see the sunlight it should definitely lead you to infer that it's daytime. Um you know the the person might still be uncertain nonetheless. So with this table here which I've always found rather interesting although much of this you know they use the word putative roles of neurotransmitters uh but but the main point here is that these are these are hypotheses regarding are there particular neurotransmitters that can onetone be linked up with the precisions on these different components in a model. um we've been loosely introduced in the preceding chapters and much more strongly we'll see them in in play in chapters 7 and 8 in the se second half of the book. Um, but the idea that that aceticoline could be what modulates the precision on likelihood, which is kind of what I'm talking about the the relationship between observations and hidden states. Um, hidden state transitions, which would be, you know, um, day goes to night and night goes to day. um day going to day is very unl you know virtually unlikely if we view this in you know a certain interval and so so having

beliefs unconsciously or not uh about how hidden states themselves transition or how they transition conditioned on action which is also how we most often see this an act of inference such as oh if you're hungry uh you have a belief that you if you eat food then you'll go from hungry to sated right? You're using an action to hit change or transition in the state of the world. Nor adrenaline um here based on just this kind of handful of study maybe a little bit more than a handful but but several studies um you know kind of point at this idea that the noradrenaline or norepinephrine could have a relationship with one's perception of how um essentially the environment changes or hidden states of the world change. um that comes up very strongly in psychiatry as well. um you know with uh my own research it's like a general finding is that individuals um with with PTSD uh tend to have kind of aberrant um uh functioning of of in production of norepinephrine and noradrenaline uh in the brain that can lead to a perception of the environment being very volatile whenever it isn't. Uh, put simply like someone having uncertainty if at any moment the hidden state of the world can go from I'm safe to I'm in danger. Um, on average those who don't have kind of the symptomatology of PTSD um tend to assume if they're in a safe situation it will continue to be safe more or less on average. Maybe not perfectly onetwoone uh which is probably for the best uh just in case something does come up but on average you're not it's it's not a volatile belief of oh dramatically it will within a blip of a moment change from a safe to unsafe situation um with PTSD it can be different right so transitions or the idea of having um sort of a precision on trans transitions in these models that's relatable to neurotrans transmission um brings up a lot of implications for thinking about things like pharmacology or at least just the brain functioning in general whether you're intending to study like medical treatments or not. Um dopamine and policies this is actually there's much more sort of evidence for this. A classic study was shortened back with Frista and others in 2014 and I briefly mentioned like dopamine being related to reward prediction errors earlier. um this this is kind of like a little bit more broadly accepted certainly within the active inference and uh literature. I'm not sure about the entirety of neuroscience, but that many others are are willing to kind of accept this sort of relationship here. Um this relates to sort of the um a colloquial word would be confidence here of saying like do you have confidence in your ability to choose actions or policies that will lead to kind of accomplishing what it is you're trying to to accomplish. Um that would kind of be a relationship to to dopamine. And you know a lot of people think of this as like a sense of reward. Um granted like excessive dopamine uh in the re-uptake system can actually lead to uh very you know on the opposite end rather problematic things such as um

you know kind of hyperactivation of the the dopamineergic system can lead someone to have rather grandiose sort of delusional beliefs um as well. So it's it's um it's found in uh schizophrenia for example, which is not to say that everyone who has schizophrenia is necessarily, you know, happy. This isn't to put a positive veilance on. It's just to say like they're they're interesting behaviors that happen with too much or too little uh of of these kinds of um quantities and serotonin um very commonly used in pharmarmacology, SSRIs and so on. um relating to, you know, potentially being the parameters on preferences or specifically interoceptive likelihood. Um I'm probably spending too much time on that, but um this it's just >> they're essentially trying to map the brain. So yeah, go ahead, Alexander. I'm sorry I've been talking a long time. >> No, it's okay. Um it's just uh obviously that figure table's adapted from a paper. Um, but it's just uh I don't know what it is about neuroscience. Um, but that is such a classic table of just like we tried to collect all of the evidence and put it in one place like these are the facts. um even if they the evidence can seem quite disperate like it's found in this region or it's like linked to this experiment or it's related to this thing and it's like it's not obvious why those pieces of evidence all build a coherent story to support that um neurotransmitter story. And so I just like I I too have made tables like this about you know a specific >> protein or protein receptor and like just trying to um yeah figure out how to uh repres represent data um and organize kind of the facts uh to build a case for um linking these together. But that is really cool that the neurotransmitters can be linked to um these specific components of the generative model. >> Sure. Yeah. Yeah. No, I I'm very with you on that. It's, you know, I appreciate that they put the word puditive here. That is like it, this is a very top down like, well, if there were a trans neurotransmitter that could be said to link up to, you know, a component of the model, this is probably what it would be, you know, based on this evidence that may or may not perfectly support the claim, but kind of points to it. So, what what I mean is it's like you they are kind of just this could be seen as an attempt to just fill out this table. Um, yeah, it's I mean this is rather significant, you know, research that's been done. I don't know if everyone on the author list in this group, well, okay, that dopamine is a bad example, but nor adrenaline. I don't know if every author here is going to look at this table and then say, "Oh, yes, this is, you know, this is what we're going for. This is what we were trying to do." It's like, um, you know, certain people were trying to study iss um, you know, those who work in pharmarmacology, I think, would be especially interested in this and also probably hope that you could, um, kind of do a onetoone um mapping of a neurotransmitter to a transition because uh to a precision rather um because it's like you know in that in there's a

you know I've been going over this paper on schizophrenia recently for example that is from an active inference perspective or framework and so they make a model you know of a of a hypothetical person and so this um this person like is pl is engaging in a behavioral task like trial by trial and we can simulate how that works and kind of how the task works and then along the way they simulate this would be the impact of using you know an SSRI and how it would change their behavior. And so if you could come up with certain kinds of behavioral criteria to say like, oh, here is a person with schizophrenia playing this game and and we get to observe them playing the game and then determine here the problematic things that happened. You know, they they thought they chose one color for a card and then it was actually a different one. They hallucinated. This sort of thing. you come up with all these criteria and then then you can create an artificial intervention by giving you know having this artificial person take an SSRI and then looking at like if it's if this is true then that means that the impact on the precision there in the model will lead to a a more normative normal behavior of them playing the game and clearly has has mitigated a lot of the sort of delusional or a delusion producing sort of symptoms in the model. And so at the end of the day we get to you know the idea would be to get into like precision pharmacology by making a purely incilico like artificial simulation there right and it's so if you could very strongly prove these sorts of things down to the degree that you could not just look at like oh is this medication going to be helpful? Yes or no? But also like how much do we need to impact the precision in the model? How much do we you know then you could really you could really fine-tune these kinds of things and maybe even fine-tune them like to each individual know exactly what dosage of what specific like chemical composition in a in a treatment would need to be. Um I love the idea of all of that. I think it's I think it's very like cool or fascinating and then I thought I think also for those who you know would prefer some kind of like pharmacology based treatment or see that as one of the few options available to them like yeah uh more precision is better uh more precision of our understanding about those things. That said, like all of these need to be all of these need to be tested more thoroughly. And yeah, to be able to make any broad statement based on this table, it's like I think the textbook authors were just trying to just kind of consolidate what they what they had to try and um you know, I I I think I think this is great for researchers who are like, "Oh, I would like to test this." It's like does serotonin if we if we collect enough behavioral data or neuro neuroimaging data or both um and then we look at people who you know are having their serotonin reuptake system modulated or not or being given a particular treatment like do we see the model that we fit to

their behavior like does something happen with their preferences matrix from there or does something happen to the precision on you know their body's ability to read itself. If it does, then maybe you do have more evidence for this kind of claim. Um, yeah, I I just I come back to this very often because I find it fascinating, but I'm always like I just need more validation. And I'm sure the authors themselves will probably agree. Um, yeah. And then this this final figure 5.5 is just like sort of trying to and this is how I describe this to to social scientists who are interested in these things but don't have any particular like strong background or potentially even interest in like fine details of neuroscience. But just the idea of being able to say like if we can map these more generally understood or at least we agree on generally like their definitions in social science of what it means to plan something or to act to have a goal versus just engage in a habit, what it means to, you know, have beliefs or predict something. like it's sort of um and then also for people who want to do any kind of cognitive modeling or or even people who don't want to do cognitive modeling but they just want to do something very hardcore um sort of robotics driven or they're trying to model an organism or something. It's like well with active inference as a framework you do if you engage in building these kinds of models you do get the ability to sort of like pinpoint as opposed to working with a total blackbox model of you know a billion parameters none of which we know exactly what they do or what they're for. um we're now able to say like oh whenever we look inside this model we do see the correlates of habits and here's why the the agent made this decision and we can look at the balance between how it evaluated you know it's its hypothetical brain evaluated its habits versus its goals and how it weighed uh certain actions based on the information gain that it thought it would get from doing something versus the the pragmatic value it thought it would get and um so there's just there's just a lot more sort of interpretability um to these models. So from a modeler's perspective, my perspective as a computational modeler, like I would love for this to be even like 80%, you know, true more or less. Like it would be wonderful to have a reasonable model that is that interpretable to where you could even go to someone who doesn't understand any of this and use kind of more a everyday action uh uh excuse me everyday language to say like oh well the the robot was planning to do this because it had these particular habits and you know it had these goals and this is how it made that decision. And so for those who are interested in things like um AI and making those like AI-driven systems to where you can actually audit the behavior of those kinds of u agents or robots or otherwise um I think that that could have a lot of sort of significant value. I think what what others are you know afraid of is a sort of you know robots start making decisions

without us really knowing or understanding why. Which means that one that leads to a lot of unpredictability on our end like what are these things going to do and then two it really harms our ability to finetune the the AI or the robots because we don't know exactly how to adjust those billion parameters that they have right in a reasonable way that is interpretable. Um so yeah and that concludes the first half of the active inference textbook and uh thank you all for joining. I uh I think I talked a little much on this one, but uh it's just uh there's a lot going on in this chapter and there's so many different ways and connections uh when when describing it or looking at it. Um so next week at the same time uh we'll jump back to cohort 8 and kind of wrap up the textbook um chapter 10 and uh and uh and Daniel will be will be doing the the morning session for that. Um he and I were going to do something different for this textbook reading session where we're going to talk more about applications of active inference and maybe even look at some actual code. Um but we know that uh one our symposium is coming up so we we weren't able to coordinate on that like terribly well given the amount of time that we had available. Um but uh more importantly we're very excited for uh Sanjivee Namoshi's textbook on active inference coming out early next year and so we're actually more ready to start talking about code whenever that comes out. Um, this textbook lends itself well to talking about mat lab code. Not everyone who attends the textbook is ready to pay for proprietary software and learn it. Um, but Sanji Nam Jooshi's textbook will have accompanying code that I believe will all be open source uh largely in Python. Um hopefully today me talking a little bit about pharmacology and trying to model psychiatric disorders counts as an application. I think it does. But uh yeah, so anyway um thank you so much folks for attending. Um appreciate you. We'll see you at the next one. Take care. >> Thank you. Thanks.